Markscheme

Mathematics: analysis and approaches
Practice question bank

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16. (a)

Consider the function $f(x) = \dfrac{5x+3}{(x+2)(x-5)}$.

Write $f(x)$ as the sum of partial fractions in the form $\dfrac{A}{x+2} + \dfrac{B}{x-5}$.

$$5x + 3 = A(x-5) + B(x+2)$$ *(M1)*

substituting $x = -2$: $-7 = -7A \Rightarrow A = 1$ **A1**

substituting $x = 5$: $28 = 7B \Rightarrow B = 4$ **A1**

[3 marks]